

RESEARCH ARTICLE

Real-Time Monitoring of Personal Protective Equipment Compliance in Surveillance Cameras

SADAM AL-AZANI¹, HAMZAH LUQMAN^{1,2}, MOTAZ ALFARRAJ^{1,2,3}, (Member, IEEE),
ALA ADDIN I. SIDIG², AYAZ H. KHAN^{1,4}, AND DINA AL-HAMED¹

¹SDAIA-KFUPM Joint Research Center for Artificial Intelligence, King Fahd University of Petroleum and Minerals, Dhahran 31261, Saudi Arabia

²Information and Computer Science Department, King Fahd University of Petroleum and Minerals, Dhahran 31261, Saudi Arabia

³Electrical Engineering Department, King Fahd University of Petroleum and Minerals, Dhahran 31261, Saudi Arabia

⁴Computer Engineering Department, King Fahd University of Petroleum and Minerals, Dhahran 31261, Saudi Arabia

Corresponding author: Sadam Al-Azani (sadam.azani@kfupm.edu.sa)

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ABSTRACT Worker safety in the industrial, construction, and energy sectors is the primary concern of governments, organizations, and project managers. Traditional monitoring systems are tedious, laborious, subjective, expensive, and inaccurate. Therefore, the development of automated and accurate systems for real-time monitoring and detecting personal protective equipment (PPE) compliance is urgently needed. This study provides valuable insights and contributions to the field of workers' safety in the industrial, construction, and energy sectors. It aims to evaluate the potential contribution of providing a robust and accurate real-time PPE compliance monitoring system using state-of-the-art deep learning techniques with real-world videos collected from surveillance cameras. Furthermore, it presents a systematic framework for developing, evaluating, and selecting the most suitable model for real-world applications, starting from data collection and ending with system deployment and maintenance. To the best of our knowledge, the collected dataset in this work is the largest PPE's dataset, with approximately 7.7 million labels representing different PPEs in 386,011 video frames collected from several surveillance cameras. The techniques adopted to generate different models and evaluated for PPE are CenterNet, Vision Transformer, and YOLOv7. Moreover, an optimized real-time inference engine is developed leveraging the high-computing power of GPUs in the system. The experimental results demonstrate that the models developed using YOLO are the most accurate and can be better generalized to detect PPE compliance in new working environments, achieving mean average precisions of 92.36% and 66.70% on *Test Set I* and *Test Set II*, respectively. In addition, they achieve real-time detection of PPE compliance.

INDEX TERMS PPE monitoring, PPE detection, multithreading, vision transformers, objects as points.

I. INTRODUCTION

Making the workplace safe is an important goal for any organization. Achieving this goal requires the industrial and construction sectors to provide and monitor the use of safety equipment by workers [1]. Personal protective equipment (PPE) is equipment worn that helps minimize exposure to hazards that can cause serious workplace injuries

and illnesses. This equipment is crucial to protect workers against health or safety risks at work. In Canada, more than 450 workers were killed in 2017, and more than 63,000 workers were injured on construction sites [2]; the majority of these cases resulted from non-compliance of workers in using PPE.

The sources of site injuries and illnesses can vary from contact with chemical and mechanical hazards to construction hazards. Accordingly, PPE can be categorized into the head, body, and respiratory PPE. Head PPE includes

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devices that help in protecting the head, face, or eyes and consists of helmets, safety glasses, face shields, and head protection. Body PPE includes vests, gloves, and safety boots, whereas respiratory PPE includes respirators and gas masks.

According to the U.S. Occupational Safety and Health Administration, non-compliance of workers in using PPE was the most violated regulations in 2017 and 2018 [3]. The traditional way of monitoring PPE compliance is human-dependent. This method is costly and difficult in most industrial and construction sites due to the large extension and complex structure of such areas; therefore, it is unfeasible and inconvenient for human operators to monitor workers who are not wearing PPE properly [4]. An automatic and real-time detection system for monitoring PPE compliance is required to address this issue. The system will help reduce the hazards in industrial and construction sites by ensuring that workers wear PPE. However, detecting workers in the video streams and monitoring PPE compliance remains a challenging task.

The advances in technology, including machine learning, hyper-performance computing, and cloud data storage, make it possible to develop computer-vision-based systems for several tasks, including image classification [5], object detection [6], and action recognition [7], among others [8]. Convolutional neural network (CNN)-based architectures, such as AlexNet [9], VGG [10], EfficientNet [11], and ResNet [12], remain the dominant models for developing most of the classification and detection applications. However, several techniques have recently been proposed using the transformer model [13] first proposed for natural language processing tasks. Due to the success of the transformer models, they have been adapted for computer vision, as vision transformers [14], which have outperformed other models in different tasks [15], [16]. To the best of our knowledge, vision transformers haven't been used for PPE, and most of the developed approaches are based on CNN models.

One of the main challenges associated with PPE compliance violation detection is the lack of annotated datasets. Few datasets have been proposed for PPE detection; most of the available datasets are either collected from web images or target only one PPE, such as a helmet or dust mask. The web-based images are usually collected from a webcam or a camera close to the workers, which is not the case in most working environments where cameras are far from workers. Chen and Demachi [17] proposed a dataset for hard hat and full-face mask images, consisting of 3,808 images collected from the web and webcam. The GDUT-HWD dataset [18] consists of 3,174 images collected from the Internet. A multi-class dataset consisting of 6,029 images of hats, dust masks, safety glasses, and safety belt images was proposed by Chen and Demachi [19]; the images were collected from the web and a SONY camera installed at six different sites.

This study presents a systematic framework, starting from data collection and ending with system deployment, for developing a system for PPE compliance monitoring in video streams using three proposed state-of-the-art models.

The main contributions of this study can be summarized as follows:

- A dataset for PPE collected from closed-circuit television (CCTV) cameras is constructed. It is annotated to cover helmets, reflective tapes, and vests. To our knowledge, it is the largest dataset for this purpose.
- Three models for PPE detection are proposed. The proposed models depend on CenterNet, Vision Transformer, and YOLOv7 for PPE detection and classification.
- A method is presented for alleviating the problem of high imbalance issue of objects' distribution at the data level.
- A comprehensive in-depth empirical analysis of several models is provided for different hypotheses, using state-of-the-art object detection techniques on two different test sets and using several evaluation measures.
- A multi-threaded inference engine is presented that is capable of processing frames at a rate of 30 frames per second (FPS) using input frames from multiple sources with the asynchronous transfer of frames from host memory to graphical processing unit (GPU) memory that hides GPU memory latency.

The remainder of this paper is organized as follows. Section II summarizes the related works, and Section III describes the proposed framework. The experimental work is presented and discussed in Section IV. A framework for system optimization and deployment is presented in Section V, and lastly, the paper is concluded in Section VI.

II. RELATED WORKS

PPE detection techniques are classified as sensor-based or vision-based [20]. Sensor-based approaches depend on sensors for tracking and monitoring workers; these sensors are attached to the worker, PPE, or both. The Global Position System (GPS) was used by Zhang et al. [21] to track workers using their helmets. A PPE checking gate equipped with radio-frequency identification (RFID) technology was designed by Kelm et al. [22]. The gate monitors PPE compliance when workers wearing an RFID pass through it. RFID was combined with the Internet of Things (IoT) by Zhang et al. [23]. The monitored PPE information was stored on the cloud to be used by mobile applications and web-based systems. The sensor-based approaches have the advantage of reducing the need for computer vision techniques to detect and segment workers and PPEs. However, attaching sensors to PPEs is costly [20]. In addition, wearing PPEs with sensors is cumbersome for real-time applications, and it is challenging to ensure that workers wear these PPEs.

The vision-based approaches are more applicable for PPE detection. These approaches utilize video cameras and do not require any special sensors, making them appropriate for real-time PPE monitoring applications. These cameras provide a raw video stream of the construction site that requires further preprocessing to be used by the following detection and classification stages of the

monitoring system. Furthermore, vision-based methods must deal with several challenges associated with raw videos, such as occlusion, low-resolution images, and changes in illumination. PPE vision-based methods are classified into classic and machine-learning techniques based on feature extraction techniques. Most of the techniques that have been proposed before 2015 can be classified as classic techniques. These approaches employ computer vision techniques to extract statistical and geometric features to detect and classify PPE objects in images. Du et al. [24] proposed a simple technique for detecting a worker's face and helmet using colors. Shrestha et al. [25] used an edge detection algorithm to detect the worker's head and helmet. Zhu et al. [26] used a histogram of oriented gradient (HOG) algorithm for extracting features from worker's head; these features were fed into support vector machines (SVM) to classify the worker as wearing a helmet or not. Rubaiyat et al. [27] used the HOG with SVM to locate the worker in the image and the Circle Hough Transform (CHT) for helmet detection. Scale-invariant features transform (SIFT) was employed for PPE detection, exhibiting better performance than other classic techniques [4].

The major improvement in PPE monitoring can be attributed to the advancement of deep learning. Several deep-learning models have been proposed for PPE detection. Pradana et al. [28] proposed a CNN model for recognizing five PPEs in a constrained environment. Wu et al. [18] used a Single Shot MultiBox Detector (SSD) to detect whether a worker is wearing a helmet or not.

Akbarzadeh et al. [29] employed two region-based CNN (R-CNN) models for PPE monitoring. The first R-CNN was used to detect workers, and the second R-CNN was used for helmet and vest detection. Fang et al. [30] employed the Faster R-CNN model for worker detection and a CNN model to classify them as wearing helmets or not. Xiong et al. [31] proposed a pose-guided anchoring framework for helmet and vest detection. The pose estimator detects the worker skeleton to be used for extracting the body part attention regions that are candidate regions of the PPE object. These regions were fed into the CNN model to learn and classify them as helmets or vests. The proposed approach was evaluated on a dataset of 305 images, and an average F1-score of 96.0% was reported. Pose information combined with an object detection algorithm was also used by Li et al. [32] for detecting helmet and safety harness hooks. A match was performed between the spatial information of the detected PPE objects and the facial features of the worker to verify if the helmet or hooks are properly worn by the worker. This approach was evaluated on a dataset of 3,208 images, and an accuracy of 94.67% was reported. Similarly, YOLOv3 and the OpenPose estimator were used by Chen and Demachi [19] for identifying the proper use of PPE in a video. The proposed approach targeted four PPE classes: helmet, mask, safety glasses, and safety belt. The proposed method was evaluated using a dataset of 6,000 images, and precision and recall of 94.47% and 83.20%, respectively, were reported.

Nath et al. [3] proposed three models for verifying the PPE compliance of workers. In the first approach, a decision tree algorithm was employed to detect workers and PPEs. The other approaches utilized YOLO and pre-trained CNN models for object detection and classification. An in-house dataset of 1,500 images with two PPEs: helmet, and vest, was used to evaluate the proposed approaches. According to the reported results, the YOLO model achieved the highest mean average precision (mAP) with 72.3%. YOLO was also used by Wang et al. [20] for detecting helmet and vest compliance. Eight detectors based on YOLO have been explored and evaluated on the Color Helmet and Vest (CHV) dataset. YOLO v5x outperformed other models with an mAP of 86.55%. However, with blurred faces, the model's performance in detecting helmets decreased by 7%. Xie et al. [33] compared the performance of YOLO with those of faster R-CNN and SSD for detecting helmets; YOLO was reported to achieve the highest mAP with 53.8%, compared with 36.3% and 40.0% for SSD and R-CNN, respectively.

Recently, the authors in [34] presented a contrastive-based YOLO7 model with an added contrast loss branch function, enhancing the YOLOv7 model's ability to distinguish and separate instances from different classes in the embedding space. It was evaluated on an extended version of the CHV dataset to cover 12 PPE objects, reporting an improvement of 2% in mAP@0.5 compared to the original YOLO7.

This study differs from the reviewed research in several key ways. It creates the largest dataset of its kind, gathered from CCTV cameras in challenging real-world environments. The study addresses various object types and tackles the issue of data imbalance. Additionally, it offers a comprehensive empirical analysis of state-of-the-art object detection techniques and includes a multi-threaded inference engine.

III. PPE MONITORING FRAMEWORK

This section describes the framework developed for PPE detection and classification. The high-level architecture of the proposed framework is depicted in Figure 1. It consists of the following modules:

- **Dataset construction:** This module defines the process of database collection and preparation. It consists of several tasks including data collection, sampling, annotation, and verification.
- **Preprocessing:** This module consists of several tasks involving the preparation of data for the forthcoming detection tasks. It includes filtering, addressing imbalance issues, and augmentation. Preprocessing steps also include the operations required for preparing the dataset for other tasks, such as model evaluation and deployment. Each of the evaluated object detection methods uses different annotation standards. CenterNet is based on the Pascal VOC annotation standard, while Vision transformer uses the COCO annotation format, and YOLOv7 is based on the YOLO format. This module also generates the required annotation format

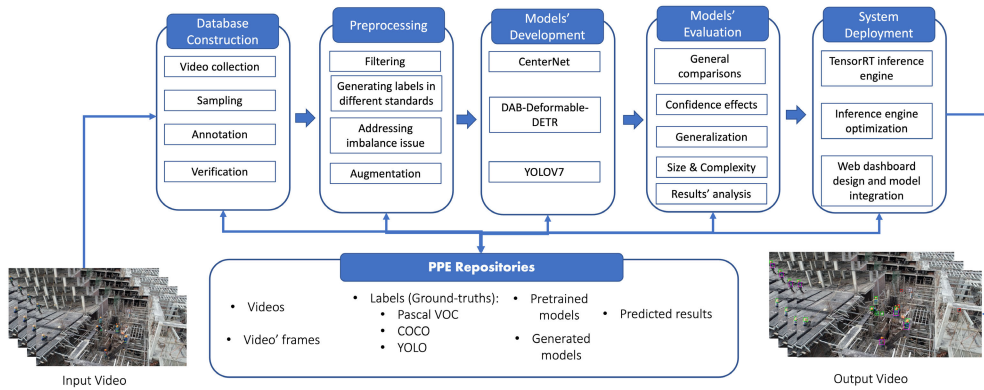


FIGURE 1. High-level architecture of the proposed framework.

TABLE 1. Statistics of the constructed database in terms of the number of labels per class, per video, the total number of labels, and the number of frames per video.

	Video1	Video2	Video3	Video4	Video5	Video 6	Video 7	Video8	Video 9	Video10	Total
Helmet	834989	987669	57073	123952	86470	175593	333744	57726	174186	218252	3049654
NoHelmet	420343	358958	9130	101333	79363	116970	7678	52785	78112	121686	1346358
Vest	311900	259502	2773	69208	47486	70150	27408	13500	40052	13542	855521
NoVest	450349	687963	63186	75979	82399	193940	303081	94607	191471	288582	2431557
All labels	2017581	2294092	132162	370472	295718	556653	671911	218618	483821	642062	7683090
# of frames	26003	45550	41538	48326	49971	53637	58094	24001	27890	11001	386011

for each technique, and includes the preparation of the implementation environments.

- **PPE models' development:** Several models have been proposed in this study for PPE detection and classification. These models depend on three state-of-the-art computer vision models: CenterNet [35], Vision Transformer [36], and YOLOv7 [37].
- **Models' evaluation:** The proposed models are trained and evaluated using the constructed dataset. Two test datasets (*Test Set I*, and *Test Set II*) are used to evaluate the models. Each set contains data collected from different CCTV cameras. Different evaluation metrics have been used to report the results of each model. These metrics are True Positive (TP), False Positive (FP), Precision, Recall, F_1 -score, Average Precision (AP), and mAP. The PASCAL VOC 2012 evaluation tool [38], adapted from MATLAB to Python by [39], is employed in this module. This tool receives the detected objects and the ground truth and reports the AP, TP, FP of each class, and mAP. We have extended this tool to report the Precision, Recall, and F_1 -score metrics. In addition, the complexity of the proposed models is evaluated in terms of the number of parameters.
- **Model's deployment:** This module defines a framework for a real-time inference engine that processes video frames from both recorded video files and camera streams. The engine is multi-threaded and it is designed to achieve a performance of 30 frames per second

(FPS) using a trained TensorRT model to detect PPE. The TensorRT model is optimized with Non-Maximum Suppression (NMS) parameters, which help to eliminate false detections. The output frames and statistics are displayed on a web-based dashboard, where they can be viewed in real-time.

A. DATASET COLLECTION AND PREPARATION

1) DATASET COLLECTION

Ten long videos have been collected from real-world environments using CCTV cameras. Most of these videos are 30 minutes in duration. Videos have been captured from different work areas, depths, and angles at different day times of day to have more variability in the captured data. The collected videos have been recorded at 30 FPS with a resolution of 1920×1080 . The videos are converted into frames using the DarkLabel tool [40]. The frames without workers are eliminated. Table 1 present the statistics of the dataset.

2) ANNOTATION

The main objective is to annotate as many PPEs appearing in the collected dataset as possible. However, it is difficult to recognize some small PPEs, such as gloves and safety shoes, due to the characteristics of the recorded videos and the distance between the CCTV cameras and workers. Therefore, the PPEs that are considered and annotated are Hamlet (hard hat), NoHelmet, Vest, and NoVest. An object in the video

frame is annotated as a *Helmet* if it is worn by a worker and the helmet fully or partially appears. This class covers all helmet colors, including blue, green, red, etc. On the other hand, an object is annotated as a *NoHelmet* if a worker is not wearing a helmet. If a worker is wearing a vest or reflective tape, the upper part of his body is annotated as *Vest*; otherwise, it is annotated as *NoVest*. Figure 2 depicts the PPEs considered as helmets with different colors and a vest (or reflective tape) from the front and back.



FIGURE 2. PPEs considered during data annotation. a) Helmets (Hard hats) with different colors, b) Vests/reflective tapes (from front and back).

DarkLabel V2.4 [40], a video labeling tool, was used for annotating PPEs in the collected videos. The PASCAL VOC annotation schema was followed in the annotation process. We also generated COCO and YOLO ground truth encodings from PASCAL VOC encoding. A guideline for annotation is first prepared, and nine annotators are involved in the annotation task. The annotations are verified carefully to ensure that the PPEs of all workers shown in the video frame are annotated, the right labels are assigned for each PPE, and the right bounding boxes are drawn around objects.

3) RELATED DATASETS

Several datasets have been proposed in the literature for PPE detection. These datasets have aided in stimulating advancements in assuring site safety. A comparison between these datasets and our proposed dataset is shown in Table 2. The comparison involves the dataset size, the number of classes, and the data source. As shown in the table, all other datasets are limited in size. The largest dataset is the safety helmet wearing detect dataset (SHWD) [41], with 7581 images. However, this dataset contains samples for helmets only, as does the GDUT Hardhat Wearing Detection (GDUT-HWD) dataset [18]. Although Pictor-v3 [3] targets helmets and vests, it has few vest samples. The dataset that contains several PPE types is CPPE-5 dataset with 6 PPEs; however, this dataset targets PPEs in the medical field which makes it inappropriate for training models for

monitoring PPE compliance in a non-medical environments, such as construction and industry. In contrast, our dataset outperformed other datasets in terms of size, with more than 386,000 images. Furthermore, almost all the samples in our dataset contain more than one worker with different PPE compliance. Moreover, all the available datasets have a close-view image whereas our dataset was collected from CCTV cameras with a wide angle using four classes helmet, no helmet, vest, and no vest.

B. PROPOSED MODELS

This section describes the three models developed in this study for PPE detection.

1) PPE-CenterNET

CenterNet is an anchor-free fast object detection model that utilizes the Object as point technique [35] for object detection. CenterNet represents an object using the geometric center of the object which is called a keypoint heatmap. The center is then used to predict the other properties of the object such as the coordinates of the bounding box, dimension, 3D extent, orientation, and pose. CenterNet is a fully CNN network that receives an input image $I \in \mathbb{R}^{W \times H \times 3}$ and generates a keypoint heatmap $\hat{Y} \in [0, 1]^{\frac{W}{R} \times \frac{H}{R} \times C}$ where R is the output and C is the number of keypoint types. In our case, $R = 4$ is the default stride in the literature and $C = 4$ equals the PPEs types (Helmet, NoHelmet, Vest, and NoVest) such that a prediction $\hat{Y}_{x,y,c} = 0$ is background while $\hat{Y}_{x,y,c} = 1$ is a detected keypoint. In order to predict an object k with bounding box $x_{min}^k, y_{min}^k, x_{max}^k, y_{max}^k$ belonging to the considered PPEs (Helmet, NoHelmet, Vest, and NoVest). The center p_k is calculated using Equation 1.

$$p_k = \left(\frac{x_{min}^k + x_{max}^k}{2}, \frac{y_{min}^k + y_{max}^k}{2} \right) \quad (1)$$

The keypoint estimator $\hat{Y}$ is used to predict all center points and the size of each detected object is regressed using Equation 2 with considering a single size prediction $\hat{S} \in \mathbb{R}^{\frac{W}{R} \times \frac{H}{R} \times 2}$ for all classes.

$$s_k = (x_{max}^k - x_{min}^k, y_{max}^k - y_{min}^k) \quad (2)$$

For each center point, a local offset $\hat{O} \in \mathbb{R}^{\frac{W}{R} \times \frac{H}{R} \times 2}$ is predicted to recover the discretization error caused by the output stride. The overall training objective is to predict key points $\hat{Y}$, offset $\hat{O}$, and size $\hat{S}$ using one single network.

$$L_{det} = L_k + \lambda_{size} L_{size} + \lambda_{off} L_{off} \quad (3)$$

where L_k is a penalty-reduced pixel-wise logistic regression with focal loss using:

$$L_k = \frac{-1}{N} \sum_{xyc} \begin{cases} (1 - \hat{Y}_{xyc})^\alpha \log(\hat{Y}_{xyc}) & \text{if } Y_{xyc} = 1 \\ (1 - Y_{xyc})^\beta (\hat{Y}_{xyc})^\alpha & \text{otherwise} \end{cases} \quad (4)$$

TABLE 2. Comparison between our dataset and other PPE datasets.

Datasets	Size (images)	PPE	Num of classes: Labels	Wide Angle
GDUT-HWD [18]	3174	Helmet	5: Yellow helmet, White helmet, Red helmet, Blue helmet, and No helmet	No
Pictor-v3 [3]	1500	Helmet, and Vest	3: Worker, helmet, and vest	No
CHV [20]	1330	Helmet and Vest	6: Worker, vest, Yellow helmet, White helmet, Red helmet, and Blue helmet.	No
SHWD [41]	7581	Helmet	2: Helmet, and No helmet	No
CPPE [31]	932	Helmet and Vest	3: Helmet, Vest, and workers	No
CPPE-5 [43]	1029	Coverall, Face Shield, Gloves, Goggles, and Mask	5: Coverall, Face Shield, Gloves, Goggles, and Mask	No
Our Dataset	386011	Helmet, Vest, and Reflective tape	4: Helmet, No helmet, Vest, and No vest.	Yes

L_{size} is the size which is trained using L1 loss, as follows:

$$L_{size} = \frac{1}{N} \sum_{k=1}^N |\hat{s}_{pk} - s_k| \quad (5)$$

L_{off} is the offset and like L_{size} it is also trained using L1 loss using Equation 6:

$$L_{off} = \frac{1}{N} \sum_p |\hat{\phi}_{\tilde{p}} - \left(\frac{p}{R} - \tilde{p}\right)| \quad (6)$$

2) PPE-TRANSFORMER

DEtection TRansformer (DETR) is the first end-to-end approach proposed for object detection using the Transformer model [43]. DETR utilizes a set of fixed, small, and learnable object queries to find the relations between the objects and the global image context via bipartite matching, an encoder-decoder architecture, and a global loss leading to parallelly predicting the final set of the objects. DETR removes the need for many hand-designed components, compared to convolutional neural architectures, such as eliminating the post-processing step of NMS. However, DETR still suffers from slow convergence and does not perform well in the detection of small objects. Consequently, many attempts have been made to solve the main issues of DETR. The proposed PPE-based transformer in this study is based on Dynamic Anchor Box and deformable attention DEtection TRansformer (DAB-deformable-DETR) [36]. DAB-deformable-DETR is based on deformable cross-attention and dynamically updated 4D anchor boxes. The deformable attention module [44] attends to a small set of key sampling points around a reference point by assigning only a fixed number of keys for each query regardless of the spatial size of the feature map as Deformable DETR. Deformable DETR is inspired by deformable convolution [45]. In contrast to the previous studies which rely solely on 2D static queries, DAB-deformable-DETR proposes dynamic queries as 4D anchor box coordinates (x, y, w, h) . The point (x, y) refers to the query center position to pool features around the center, and (w, h) refers to the anchor box size to modulate the cross-attention map. The queries are updated dynamically layer by layer. Introducing queries considering both size and position can enrich the cross-attention module. Inspired by [43], the

high-level structure of our PPE-Transformer is composed of a backbone, transformer encoder, transformer decoder, and prediction heads. The backbone module receives a video frame (image) $x_{img} \in \mathbb{R}^{3 \times H_0 \times W_0}$ as an input and extracts the spatial features, a lower-resolution activation map $f \in \mathbb{R}^{C \times H \times W}$, where the typical values are $C = 2048$. The extracted features are then fed into the Transformer encoder for refinement. The encoder module comprises normalization layers, skip connections, a multi-head self-attention module, and a feed-forward network (FFN). It first generates a new feature map $z_0 \in \mathbb{R}^{d \times H \times W}$ such that $d < C$ using 1×1 convolution filter. It then transforms z_0 into a token of one dimension $d \times HW$ by adding spatial position embedding to queries and keys at each attention layer. The generated embeddings are fed into the transformer decoder, in which the object queries (positional encodings) are learned and added to each attention layer. The transformer decoder transforms the N object queries into the corresponding embeddings. The embeddings are fed into the FFN to be decoded independently into box coordinates and detecting PPE labels.

3) PPE-YOLO

You-only-look-once (YOLO) is an object detection model that can detect objects in images and videos. It works by dividing an input image into a grid and using the grid to determine which objects appear within each section of the grid. It also uses a neural network to classify objects. YOLO is fast and accurate, making it ideal for real-time applications. It is also robust, as it can handle changes in the environment, such as moving objects. The main framework of YOLOv7 comprises three main components: Backbone, Neck, and Head. The Backbone is a CNN-based architecture that receives an input image and extracts the features at different granularities. The extended efficient layer aggregation Network (E-ELAN), as an extended version of the efficient layer aggregation network (ELAN) was proposed to be the Backbone architecture of YOLOv7, which is composed of several operations including expand, shuffle, and merge. The extracted features are then fed into the Neck component, which is the intermediate layer between the Backbone and the Head. The neck is based on the featured pyramid network (FPN), which combines ConvNet

layer representations and generates the feature pyramid. The generated features are then fed into the Head layer to predict the bounding boxes and the classes, trained using three loss functions: cross-entropy loss, for class prediction, L1 loss for bounding box prediction, and objectness loss for object existence detection. In contrast to vision transformer-based object detection techniques, YOLOv7 relies on the NMS technique as a postprocessing step to eliminate more than one candidate object proposals for classification.

IV. EXPERIMENTS AND RESULTS

A. DATASET

We used the dataset compiled in this study for training and evaluating the proposed models. The dataset is divided into training, validation, and testing sets. The test set contains videos captured in an environment different than the training set. Two sets have been used in this work for evaluating the proposed models, *Test set I* and *Test set II* which contain frames extracted from Video 2 and Video 7, respectively. The remaining videos are used for model training and validation. More information about the dataset can be found in Section III-A. To alleviate the high-imbalanced class issue in the training set, all frames within the four considered classes (NoHelmet, Helmet, NoVest, and Vest) are included for the training phase such that the ratio of Helmet to NoHelmet and the ratio of NoVest to Vest is ≤ 3 . The first 90% of the extracted frames from each video are selected for training, and the last 10% are used for validation. Table 3 presents the statistics of the dataset used for the models' development and evaluation.

B. EXPERIMENTAL SETUP

The models were trained using the default parameters as the best parameters, performing some modifications based on the results obtained on our validation sets. For the PPE-CenterNet, the model was trained on an input image of 1024×1024 , applying image cropping, brightness, and random scaling as data augmentation. We used a $5e-4$ learning rate for 100 epochs with a learning rate dropped 10x at 50 and 90 epochs and a batch size of 16, ReLU activation function, and ADAM optimizer. For the PPE-Transformer, the model was trained on input frames of 1920×1080 . Noise light, brightness, contrast, and horizontal flip were used as data augmentation. We used $1e-4$ learning rate for 100 epochs with a learning rate dropped 10x at 40 and 80 epochs, ReLU activation function, and ADAM optimizer. The model was trained using eight attention heads, 300 query slots, six encoding layers, and six decoding layers. ResNet50 was used as a feature extraction backbone. The PPE-YOLO was trained using an input image of 1920×1920 , applying image cropping, brightness, random scaling, hue, horizontal flip, saturation, and translation as online augmentation. The momentum factor, learning rate, and decay rate are set to 0.937, 0.01, and $5e-4$, respectively.

C. EVALUATION METRICS

We consider Recall, Precision, F_1 , and AP evaluation metrics to evaluate the proposed models. It should be emphasized that the evaluation metrics related to object detection are slightly different from standard classification tasks [46]. All models are evaluated using an intersection over union (IoU) of 0.5. IoU represents the overlap between the ground truth bounding box (B_g) and the predicted bounding box (B_p). It is used with two different confidence thresholds, 0.0 and 0.3. The confidence score threshold is used in object detection tasks to filter out false positives and ensure that a predicted bounding box has a specific minimum score [47]. The first confidence threshold of 0.0 indicates that all detected objects with confidences of > 0.0 will be considered, whereas the second confidence threshold of 0.3 indicates that only objects detected with confidences of > 0.3 will be considered. The confidence threshold of 0.0 is the default threshold used for evaluating object detection models, while the confidence threshold of 0.3 is the recommended threshold for the inference phase [36], [44]; it is also selected based on our investigation of different confidences on the validation set. Precision and Recall are calculated for each prediction based on a sorted list of detections to calculate the mAP. The Precision and Recall are calculated using the IoU value for a given IoU threshold. Recall, Precision, and F_1 metrics are calculated based on the IoU, where True Positive (TP) refers to the correct detection of ground-truth bounding boxes, False Positive (FP) refers to incorrect detection of nonexistent objects or misplaced detection of existing objects; False Negative (FN) refers to undetected ground-truth bounding boxes. The mAP is the mean of APs of all classes. The AP is calculated for each class as follows:

$$AP : \sum_{i=1}^n p(i) \Delta r(i) \quad (7)$$

where n is the total number of detections, i is the rank of singular detection within the list of sorted detections, $p(i)$ is the cumulative precision from first to the i^{th} detection, and $\Delta r(i)$ denotes the variation in Recall from the $(i-1)^{th}$ to the i^{th} detection.

D. RESULTS

Table 4 presents the obtained results using the PPE-CenterNet model on *Test Set I* and *Test Set II*. For *Test Set I*, the highest mAP of 64.03 is obtained when considering all detected objects (i.e., confidence threshold of 0.0) while the highest overall average of Precision and F_1 are obtained for the detections with a confidence threshold of 0.3. There is an insignificant difference between the Recall reported for all detections (confidence threshold of 0.0), and that for objects detected with a confidence threshold of 0.3. It can be seen that the Helmet class is the most accurate detected class, followed by the Vest class with APs of 69.42% and 66.83%, respectively. However, the lowest results are obtained for the NoVest class in nearly all cases. PPE-CenterNet demonstrates

TABLE 3. Statistics of the dataset used for the models' development and evaluation.

	# of frames	# of objects	NoHelmet	Helmet	NoVest	Vest
Training	95100	2634324	584859	954069	726220	369176
Validation	10580	363842	87107	132965	102107	41663
<i>Test Set I</i>	45550	2294092	358958	987669	687963	259502
<i>Test Set II</i>	58094	671911	7678	333744	303081	27408
All Test Sets	103644	2966003	366635	1321412	991042	286914
Total	209324	5964169	1038601	2408446	1819369	697753

poor results for *Test Set II*, as shown in the bottom part of Table 4, reporting mAPs of 21.59% in the best case when considering all detected objects and 20.72% when considering only the objects with a confidence threshold of 0.3. NoHelmet is the most challenging class, while Helmet is the most accurate detected class with an AP of 41.91% for objects detected with a confidence threshold of 0.0 and an AP of 20.72% when just considering the objects with a confidence threshold of 0.3.

The obtained results using the DAB-Deformable-DETR model are presented in Table 5. Similar to the CenterNET-based model, the DAB-Deformable-DETR achieved higher results with *Test Set I* compared with *Test Set II*. For *Test Set I*, the DAB-Deformable-DETR model achieves an mAP of 72.91% for all detected objects and achieves an mAP of 68.05% for objects detected with a confidence threshold of 0.3. The Helmet class is the most accurate detected class with an AP of 81.92% for all detected objects and an AP of 75.16% for objects detected with a confidence threshold of 0.3. The lowest results for *Test Set I* are obtained for the NoHelmet class with APs of 62.34% and 59.35% for detected objects with confidence thresholds of 0.0 and 0.3, respectively. For *Test Set II*, the transformer-based model achieved an mAP of 58.32% for all detections (confidence threshold of 0.0) and 53.23% for the detected objects with a confidence threshold of 0.3. The Helmet is the most accurate class with APs of 86.78% and 79.24% for detected objects with confidence thresholds of 0.0 and 0.3, respectively. In addition, the model performs well for the NoVest and the Vest classes with reporting APs of 79.69% and 63.54%, respectively for detections with a confidence threshold of 0.0. However, the model can't perfectly detect objects belonging to the NoHelmet class, reporting APs of 3.28% and 2.38% for detected objects with confidence thresholds of 0.0 and 0.3, respectively.

The results obtained using the PPE-YOLO model are presented in Table 6 for *Test Set I* and *Test Set II*. For *Test Set I*, PPE-YOLO reports the highest overall mAPs of 92.36% in case of considering objects detected with a confidence threshold of 0.0, while achieving mAP of 85.05% when just considering the detected objects with a confidence threshold of 0.3. The Helmet is the most accurate detected class with APs of 96.79% and 93.10% for detected objects with confidences thresholds of 0.0 and 0.3, respectively. The lowest results for *Test Set I* are obtained for the NoVest class with an AP of 87.70% for all detected objects (confidence

threshold of 0.0) and for the Vest class with an AP of 84.04% for the detected objects with a confidence threshold of 0.3. For *Test Set II*, PPE-YOLO also achieves the highest mAP of 66.70% for all detected objects and mAP of 58.42% for the detected objects with a confidence threshold of 0.3. The Helmet is the most accurate class with APs of 96.05% and 92.61% for detections with confidence thresholds of 0.0 and 0.3, respectively. In addition, the model performs well for the NoVest and the Vest classes with reporting APs of 85.30% and 74.11%, respectively for objects detected with a confidence threshold of 0.0. The model also can detect the Vest and the NoVest objects with APs of 73.53% and 62.61%, respectively for the detected objects with a confidence threshold of 0.3. However, the model cannot accurately detect objects belonging to the NoHelmet class, reporting APs of 11.34% and 4.86% for objects detected with confidence thresholds of 0.0 and 0.3, respectively.

E. DISCUSSION

As shown in Tables 4, 5, and 6, PPE-YOLO obtained the highest AP for all classes and the highest overall average mAP for all cases with *Test Set I*, as depicted in Figure 3. The lowest mAP is reported using CenterNet. However, PPE-CenterNet achieves better AP than PPE-DAB-Deformable-DETR in the case of detecting the NoHelmet class for all the detected objects (with a confidence threshold of 0.0) and for detecting objects with a confidence threshold of 0.3. It also performs better than PPE-DAB-Deformable-DETR for the Helmet class in the case of detecting objects with a confidence threshold of 0.3. In addition, PPE-YOLO reports the highest Recall score in all cases. However, PPE-CenterNet achieves the highest Precision and F_1 in the case of detecting all objects (with a confidence threshold of 0.0), as depicted in Figure 4. Even though PPE-CenterNet reports the highest results, PPE-YOLO remains the best in terms of TP and FP. This is why PPE-YOLO reports the highest overall mAPs. For *Test Set II*, PPE-YOLO also reports the highest mAPs for all cases, as depicted in Figure 5, except in the case of the Vest class detected with a confidence threshold of 0.3 in which, PPE-DAB-Deformable-DETR performs better. For evaluation measures other than AP, the best performance is achieved using PPE-YOLO for all objects detected (with a confidence threshold of 0.0), while PPE-DAB-Deformable-DETR reports the highest overall Precision, Recall, and F_1 in case of detecting objects with a confidence threshold of 0.3, as shown in Figure 6.

TABLE 4. Results of PPE-CenterNet, GT: Ground Truth. The upper rows present the results for *Test Set I*, while the bottom rows present the results for *Test Set II*. The left and right parts show the results for objects detected with confidence thresholds of 0.0 and 0.3, respectively.

		Confidence > 0.0								Confidence > 0.3					
		GT	TP	FP	Prec	Recall	F1	AP		TP	FP	Prec	Recall	F1	AP
<i>Test Set I</i>	NoHelmet	358957	244225	50945	82.74	68.04	74.67	65.71		243820	27642	89.82	67.92	77.35	65.61
	Helmet	987668	696161	46708	93.71	70.49	80.46	69.42		691558	22237	96.88	70.02	81.29	68.97
	NoVest	687961	398975	98227	80.24	57.99	67.33	54.16		397082	60057	86.86	57.72	69.35	53.93
	Vest	259506	182597	45138	80.18	70.36	74.95	66.83		181757	34858	83.91	70.04	76.35	66.56
	All	2294092	1521958	241018	84.22	66.72	74.35	64.03		1514217	144794	89.37	66.43	76.09	63.77
<i>Test Set II</i>	NoHelmet	7678	164	167708	0.1	2.14	0.19	0.04		100	43191	0.23	1.3	0.39	0.04
	Helmet	333744	171540	415738	29.21	51.4	37.25	41.91		151347	153160	49.7	45.35	47.43	39.5
	NoVest	303081	79717	518179	13.33	26.3	17.7	18.22		69089	194221	26.24	22.8	24.4	17.49
	Vest	27408	11898	281589	4.05	43.41	7.42	26.19		10813	71489	13.14	39.45	19.71	25.86
	All	671911	263155	1215506	11.67	30.81	15.64	21.59		231349	462061	22.33	27.22	22.98	20.72

TABLE 5. Results of vision Transformer PPE-DAB-Deformable-DETR, GT: Ground Truth. The upper rows present the results for *Test Set I*, while the bottom rows present the results of *Test Set II*. The left part shows the results for the detected objects with a confidence threshold of 0.0, while the right part is for the detected objects with a confidence threshold of 0.3, respectively.

		Confidence > 0.0								Confidence > 0.3					
		GT	TP	FP	Prec	Recall	F1	AP		TP	FP	Prec	Recall	F1	AP
<i>Tset Set I</i>	Nohelmet	358957	259949	767290	25.31	72.42	37.51	62.34		245134	54900	81.70	68.29	74.40	59.35
	Helmet	987668	849393	866377	49.51	86.00	62.84	81.92		770306	47217	94.22	77.99	85.34	75.16
	NoVest	687961	518170	516906	50.06	75.32	60.15	70.16		470697	63095	88.18	68.42	77.05	64.72
	Vest	259506	213278	563637	27.45	82.19	41.16	77.21		198456	28233	87.55	76.47	81.64	72.97
	All	2294092	1840790	2714210	38.08	78.98	50.41	72.91		1684593	193445	87.91	72.79	79.61	68.05
<i>Test Set II</i>	Nohelmet	7678	5580	1962452	0.28	72.68	0.56	3.28		3235	58706	5.22	42.13	9.29	2.38
	Helmet	333744	309021	1447906	17.59	92.59	29.56	86.78		270421	13029	95.40	81.03	87.63	79.24
	NoVest	303081	264202	696231	27.51	87.17	41.82	79.69		220716	29034	88.37	72.82	79.85	69.14
	Vest	27408	25785	1098223	2.29	94.08	4.48	63.54		24199	39080	38.24	88.29	53.37	62.16
	All	671911	604588	5204812	11.92	86.63	19.11	58.32		518571	139849	56.81	71.07	57.54	53.23

TABLE 6. Results of PPE-YOLO, GT: Ground Truth. The upper rows present the results for *Test Set I*, while the bottom rows present the results for *Test Set II*. The left and right parts show the results for the detected objects with confidence thresholds of 0.0 and 0.3, respectively.

		Confidence score > 0.0								Confidence score > 0.3					
		GT	TP	FP	Prec	Recall	F1	AP		TP	FP	Prec	Recall	F1	AP
<i>Test Set I</i>	NoHelmet	358957	345771	647648	34.81	96.33	51.14	93.78		320890	17785	94.75	89.40	91.99	87.94
	Helmet	987668	965792	466497	67.43	97.79	79.82	96.79		926847	24442	97.43	93.84	95.60	93.10
	NoVest	687961	644194	1044031	38.16	93.64	54.22	87.70		545892	51016	91.45	79.35	84.97	84.04
	Vest	259506	250075	450450	35.70	96.37	52.10	91.17		224788	32145	87.49	86.62	87.05	76.51
	All	2294092	2205832	2608626	44.02	96.03	59.32	92.36		2018417	125388	92.78	87.30	89.91	85.40
<i>Test Set II</i>	NoHelmet	7678	7507	824543	0.90	97.77	0.90	11.34		2370	14893	13.73	30.87	19.00	4.86
	Helmet	333744	328206	992854	24.84	98.34	24.84	96.05		312888	26478	92.20	93.75	92.97	92.61
	NoVest	303081	287231	1193094	19.40	94.77	19.40	85.30		192622	11200	94.51	63.55	76.00	62.68
	Vest	27408	26720	770282	3.35	97.49	3.35	74.11		25546	76852	24.95	93.21	39.36	73.53
	All	671911	649664	3780773	12.13	97.09	20.04	66.70		533426	129423	56.34	70.34	56.83	58.42

The highest results in terms of AP score are obtained when considering all detected objects with a confidence threshold of 0.0. That means the reported APs with a confidence threshold of 0.0 is larger than the reported APs

with a confidence threshold of 0.3, in all cases except for the NoHelmet class for the *Test Set II* using PPE-CenterNet, which is not be affected by the confidence. However, the drop in terms of APs is either significant

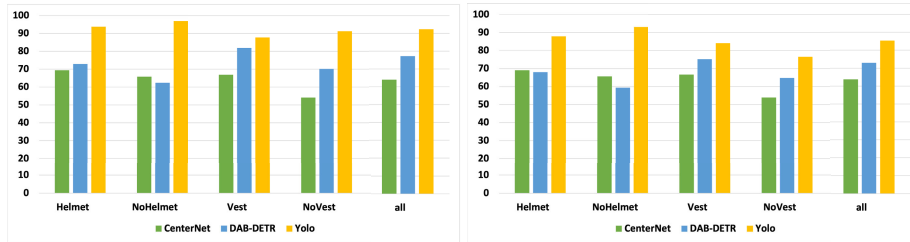


FIGURE 3. AP for each class and overall mAP of *Test Set I* with left) confidence score > 0.0, right) confidence score > 0.3.

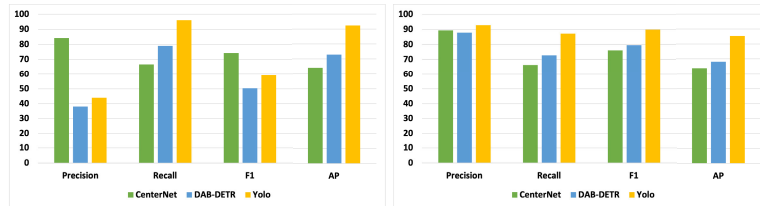


FIGURE 4. Overall average Precision, Recall, F1, and AP of *Test Set I* with left) confidence score > 0.0, right) confidence score > 0.3.

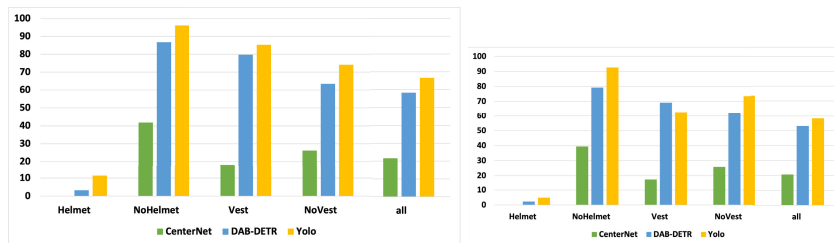


FIGURE 5. AP for each class and the overall mAP *Test Set II* with left) confidence score > 0.0, right) confidence score > 0.3.

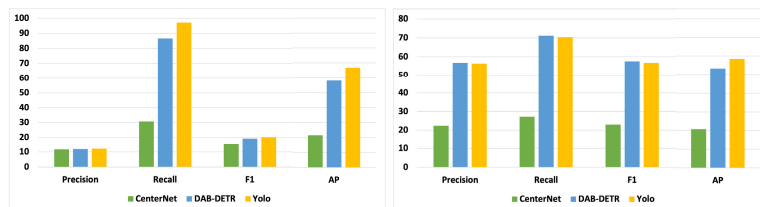


FIGURE 6. Overall average Precision, Recall, F1, and mAP of *Test Set II* with left) confidence score > 0.0, right) confidence score > 0.3.

or insignificant. As can be seen in Table 4, there is an insignificant drop in AP using the evaluated confidence thresholds for the CenterNet-based model. This is affected by the FP evaluation measure reported using the evaluated object detection techniques in Tables 4, 5, and 6, such that the FPs of transformer-based models are much higher than the FPs of CenterNet-based models, for all cases which are followed by YOLO-based models. Conversely, the performance of the generated models in terms of Precision significantly increases with the confidence threshold of 0.3 compared with the confidence threshold of 0.0, in all cases. The overall average improvements of Precision score are approximately 5% and

11% in the case of the CenterNet-based model for *Test Set I* and *Test Set II*, respectively, and approximately 50% in the case of the Transformer-based model for both *Test Set I* and *Test Set II*. Similarly, the Precision score is also significantly improved by around 50% for *Test Set I* and by 45% for *Test Set II*.

Like the AP score, the Recall score decreases for both test datasets when increasing the confidence. However, the drops are more significant for the Transformer-based and YOLO-based models than the CenterNet-based models. The overall average of Recall obtained using the CenterNet-based model appears stable for *Test Set I* and decreases from

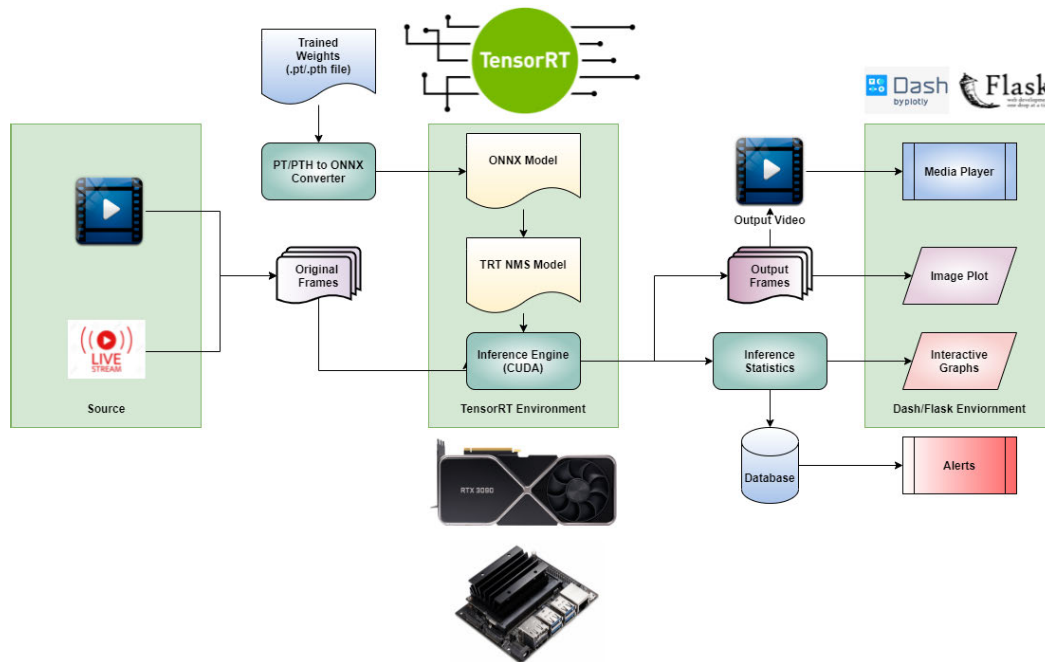


FIGURE 7. Real-time inference engine for PPE monitoring.

30.81% to 27.22% when increasing the confidence. On the other hand, the obtained overall average of Recall using the Transformer-based model notably decreases from 78.98% to 72.79% for *Test Set I* and from 86.63% to 71.07% for *Test Set II*. This is also the case for YOLO-based models, in which the reported overall average of Recall dramatically decreases by 10% for *Test Set I* and by approximately 27% for *Test Set II*. Because the Precision score is dramatically improved when increasing the confidence compared to the drop of Recall, F_1 also improves when increasing the confidence for all classes and for both test sets. The improvements in F_1 score are much more significant for Transformer-based and YOLO-based models, while it is not as significant in the case of CenterNet-based models, especially for *Test Set I*. That means anchor-free models are insensitive to the confidence score, and most of the objects are detected with high confidence scores. However, PPE-DAB-Deformable-DETR and PPE-YOLO are affected by the confidence scores, and many objects are detected with low confidence scores.

The proposed models have been evaluated using two different test sets to evaluate the generalizability of the evaluated object detection techniques. Both test sets are from videos different from those used for training and validation. However, *Test Set I* belongs to Video 2 which is from the same angle as Video 1 at different times, while *Test Set II* is completely recorded from a different angle and in a different work environment which is more challenging. This is why the results obtained from *Test Set I* are much higher than those obtained from *Test Set II* for all assessed techniques. As can be seen in Tables 4, 5, and 6, the highest results in terms of Precision, Recall, F_1 , and mAP are obtained using PPE-YOLO, while the lowest results are obtained using

PPE-CenterNet. For the Helmet and the NoVest classes, PPE-DAB-Deformable-DETR reports higher results for the most challenging dataset (Test Set II) than Test Set I, especially for AP and Recall in case of all detections with the confidence threshold of 0.0 and for all evaluation measures in case of detections with the confidence threshold of 0.3. It is also important to compare the complexity of the generated models in terms of the number of parameters. The number of parameters of the generated models using PPE-CenterNet, PPE-DAB-Deformable-DETR, and PPE-YOLO is approximately 5 million, 47 million, and 36.5 million, respectively.

V. REAL-TIME SYSTEM ARCHITECTURE

We have developed a multi-threaded inference engine for detecting the PPE used by the workers on the construction site targeting the performance of 30 FPS. Figure 7 illustrates the overall system architecture, taking original frames coming directly from the camera or extracted from the input video file, processing these frames using the inference engine, and then generating the output frames as well as statistics for PPE detection to be shown in a web-based dashboard developed in this study.

A. INPUT SOURCES

The system architecture supports both the recorded video file and the camera stream as a direct source of the original frames extracted for inferencing. These input frames are pre-processed depending on the input source (video or camera) in a separate thread; the processed frames are then saved in a shared queue as input (IQueue) to the inference engine.

B. TensorRT INFERENCE ENGINE AND OPTIMIZATIONS

The trained model is loaded in a TensorRT engine execution context on the GPU as a non-default CUDA stream. This requires the model to be converted to a TensorRT optimized model from the original PyTorch in two steps:

- 1) The PyTorch model checkpoint is converted to the ONNX model including the NMS parameters; this will eliminate the boxes which have low confidence or do not have objects and keep the most representative ones.
- 2) The ONNX model is then converted to a TensorRT model with the NMS plugin enabled for inference result optimizations.

The TensorRT model reads the original frames from the IQueue one by one, asynchronously transferring the frame from host memory to GPU memory while running model inference on each frame. This will overlap the model inferencing with the memory transfer to hide the GPU memory latency. The output frames are then saved in another shared queue as output (OQueue). Additionally, a text file is generated for each output frame with the final boxes, scores, and detected class of the PPE object. This file is used by the dashboard for overall stats visualization.

The TensorRT inference engine is capable of processing frames at the rate of 338.65 FPS on an NVIDIA RTX A6000 GPU. Table 7 shows the complete GPU specifications.

TABLE 7. GPU specifications.

CUDA Driver/Runtime Version	11.7
CUDA Capability	8.6
Global Memory	48 GB
CUDA Cores	10752 (84 MP, 128 CUDA Cores/MP)
Shared Memory	48 KB/block, 100 KB/MP
Number of Threads	1024/block, 1536/MP

However, we observed a significant performance bottleneck in reading a frame, performing model inferencing, and generating the output frame in the sequence, limiting the overall FPS to 8.6. Table 8 shows the rates of reading frames, model inferencing, and generating output frames.

TABLE 8. Performance bottleneck in the inferencing process.

	FPS
Input Frames	28.54
Inferencing	338.65
Output Frames	122.08

Our multi-threaded model inference architecture overlaps the three steps (input, inferencing, output) to increase the throughput of the overall system. Furthermore, it overlaps the communication with computation on the GPU by asynchronous memory transfers between the host system memory and the GPU memory. This improves the overall performance to 28.65 FPS with minimal system resource utilization. Table 9 shows the overall resource utilization to process a complete video stream of approximately 30 secs.

TABLE 9. Overall system resource utilization.

CPU Maximum	10.70%
CPU Memory	12.07 GB
GPU Maximum	8%
GPU Memory	1.44 GB

C. WEB VISUALIZATION

A web-based dashboard was developed in this study to visualize and show the statistics of the detected PPEs. The dashboard reads the output frames from the OQueue and performs base64 encoding to display it as an image in the Dash/Flask HTML Image component. Additionally, it compiles the statistics from the output text files and stores these statistics in the database to generate real-time visualization on the web dashboard.

VI. CONCLUSION

This study presents a framework for developing a real-time system for PPE detection and classification. It starts with constructing a large dataset collected from CCTV cameras. It also evaluates, for the first time, three state-of-the-art object detection models: CenterNet, Vision transformer (DAB-Deformable-DETR), and YOLOv7. The generated models were evaluated using two unseen test sets. The *Test Set I* was collected from one of the cameras used to collect the training set but at different working times, while the *Test Set II* was collected from a different camera at different work environment, as an out-of-distribution test set. The experimental results reveal that the PPE-YOLO model achieved the highest performance, reporting an mAP of 92.36% for *Test Set I* and an mAP of 66.70% for *Test Set II* which is followed by DAB-Deformable-DETR. The lowest results were reported using CenterNet, in nearly all cases. On the other hand, CenterNet was able to detect objects with higher confidence than DAB-Deformable-DETR and YOLOv7. This study also provides evidence that it is insufficient to evaluate the object detection techniques using only a single evaluation measure. In addition, this study provides a framework for a real-time multi-threaded inference engine for detecting PPE with a frame rate of 30 FPS.

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SADAM AL-AZANI received the B.Sc. degree (Hons.) in computer science from Dhamar University, Thamar, Yemen, in 2004, and the M.Sc. and Ph.D. degrees in computer science from the King Fahd University of Petroleum and Minerals (KFUPM), Dhahran, Saudi Arabia, in 2014 and 2019, respectively. He is a Founding Research Scientist with SDAIA-KFUPM Joint Research Center for Artificial Intelligence (JRC-AI), KFUPM. He has several publications and patents in his research areas. He also served as a PI and a Co-I for several funded research projects, demonstrating a strong leadership role and collaborative expertise in advancing various research initiatives. His research interests include artificial intelligence, multimodal learning, natural language processing, computer vision, generative AI, video analytics, and social network analysis. He has served as a program committee member and a reviewer for a number of international conferences and journals.



HAMZAH LUQMAN received the master's and Ph.D. degrees in computer science from the King Fahd University of Petroleum and Minerals (KFUPM), Saudi Arabia, in 2013 and 2018, respectively. He is an Associate Professor with the Department of Information and Computer Science, KFUPM. His primary research interests include computer vision and Arabic natural language processing. He has been granted funding for several projects from various organizations and has several publications and patents in his research areas. His papers have been published in top journals and conferences, such as *Pattern Recognition*, *Information Processing and Management*, *Engineering Applications of Artificial Intelligence*, CVPR, WACV, and EMNLP.



MOTAZ ALFARRAJ (Member, IEEE) received the B.Sc. degree (Hons.) in electrical engineering from the King Fahd University of Petroleum and Minerals (KFUPM), Dhahran, Saudi Arabia, in 2013, and the M.S. and Ph.D. degrees in electrical and computer engineering with a minor in mathematics from Georgia Institute of Technology, Atlanta, GA, USA, in 2015 and 2019, respectively. He is an Assistant Professor with the Electrical Engineering Department and the Chairman of the Information and Computer Science Department, KFUPM. He was the Founding Director of SDAIA-KFUPM Joint Research Center for Artificial Intelligence, KFUPM, from 2021 to 2024. His research interests include machine learning, deep learning, computer vision, and image processing. His research focuses on the integration of physics in data-driven systems to enable effective learning from noisy or unannotated data for applications in oil and gas exploration and production. He is a member of the IEEE Signal Processing Society (SPS), the Society of Exploration Geophysicists (SEG), and the Society of Petroleum Engineers (SPE). He was a recipient of the Outstanding Research Award from the Center for Signal and Information Processing (CSIP) at the Georgia Institute of Technology in Spring 2019. He is currently serving as the Chapter Chair for IEEE Signal Processing Society, Saudi Arabia Section.

ALA ADDIN I. SIDIG received the Ph.D. degree in computer science and engineering from the King Fahd University of Petroleum and Minerals (KFUPM). He is currently an Assistant Professor with KFUPM. His research interests include computer vision, natural language processing, and machine learning. He has published several papers in reputed journals and conferences.



AYAZ H. KHAN received the bachelor's degree in computer science and information technology, the M.S. degree in computer science, and the Ph.D. degree in computer science and engineering from the King Fahd University of Petroleum and Minerals (KFUPM), Saudi Arabia. He is an Assistant Professor with the Computer Engineering Department, KFUPM, and also holding a research fellow position with the SDAIA-KFUPM Joint Research Center for Artificial Intelligence. His academic inclinations lie in high performance computing and deep learning, with over 25 publications to his name and a notable book on parallel processing. Additionally, he possesses several professional certifications.



DINA AL-HAMED is a Distinguished Graduate (Hons.) in artificial intelligence from the College of Computer Science, Imam Abdulrahman Bin Faisal University. With a solid foundation built with the King Fahd University of Petroleum and Minerals, she has demonstrated expertise in a wide range of AI technologies and applications. Her research interests include all fields of AI, with a particular focus on computer vision and natural language processing.

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